Bridge C review

Bridge C review: missed variables, patterns, correlations, contradictions

2026-10-24, 8:24 PM CDT (America/Chicago). Read-only review of BRIDGE_C_35/order20, order27–37 and MAPS_CANDIDATES. I ran no new numerics or hashed tests and edited no existing file. `papers/RTA_FOUNDATION_BOUND/` does not exist under `/workspace/quantum-comp/papers/`, so I could not check the v10 paper notes. I found no running order37 process, and order37/RESULTS.md is complete (RG34 FAIL).

Executive findings

- **Most serious tension.** HUNT_ALPHA1 (19:35) says *"There is no written mod-8 indexing of the 105 companions or of the T-phases k/9800"* and *"Maps stay paused"*. PATTERN_BNU_RR (PBR2 "140·T-phase mod 8", 19:52), PATTERN_S_MARKOV and PATTERN_ST_MARKOV ran after that and use exactly this kind of indexing. Only PATTERN_FIBRE_RR and PATTERN_QNU_RR carry an authorization line ("following the mathematical patterns"). SIGNED, BNU, S_MARKOV and ST_MARKOV have none.
- **Order34 says something later orders refute.** order34 RESULTS/NOTES: *"its miss is the L_B leakage"*, *"0.3% miss is L_B leakage outside the companions"*. order35 (by characteristic) and order36 (by span) both FAIL that reading: rel. diff 5.03 and 17.9. Order34 has no forward note.
- **Order30's "generic" near-law is refuted by order31, but order30 is not annotated.** order30 RESULTS still says *"looks generic to unimodular cone corners"*. order31 RG25a gives 26/30 FAIL and an e1-only reading.
- **Order31 has a count inconsistency.** The text says *"The four that satisfy it are exactly the cones with edge ray (1,0)"*, but the table lists 5 cones, K included. NOTES says "5/5 vs 0/26" (31 cones). RG25a is defined over 30 cones.
- **Three different maps give one vector.** PSM1 push = PBR2 = PST1 = (8/35, 11/70, 0, 9/70, 19/70, 3/35, 0, 9/70), and all have TV 71/140. That is one finding, not three independent FAILs.
- **The fibre parity structure is clean and nobody has tested it.** C35 (type (1,1)) lives only on odd fibres {1,3,5,7}. Class (0,0) lives only on {0,4}. MF2 puts total mass exactly ½ on odd fibres. Residual empties {3,7} = 3 mod 4; T-phase/Markov empties {2,6} = 2 mod 4. MF2 folded mod 4 is (¼, ½, ¼, 0).
- **There is a 105 + 35 = 140 coincidence across the geometric and map sides.** The ⟨γ,h⟩ character ranks (order27) are 36 + 24 + 18 + 27 = 105, the companion count,

and the (+,+) rank is 36 = dim V. The ST-Markov classes split 140 = C105 + C35. No package relates the two.

- **The near-law ranks by the norm of the edge.** e1 (Q = 1): r ≈ 0.003. e2 (Q = 7/2): r ≈ 0.022. Interior: 0.04–0.17 (order34). Q(edge) has never been a preregistered predictor.
- **r shrinks as the grid fills, and Im $\hat{\kappa}$ shrinks with it.** P7: 0.0585 / Im $\hat{\kappa}$ −0.0229. G48: 0.019 / −0.0043. Full 216 grid: 0.00305 / −0.00034. Im $\hat{\kappa}$ is always negative and its ratio to r falls (0.39 → 0.23 → 0.11). Im $\hat{\kappa}$ is never used as a variable.
- **The P7 value matches the corner value.** r(C2b on P7) = 0.05853 (order35) equals r(CO, w = −1, full grid) = 0.0585 (order29/31). It may be a coincidence. Check it before using it.
- **Order35 repeats one number.** Outside share ‖S_out‖/‖S‖ = 0.926 equals the H1 perpendicular share 0.926. That may be correct by construction, but it is unexplained.
- **The S/X out-of-span fractions are equal.** 0.3678 vs 0.3677 (order36), well below any noise level. This is the strongest unexplained correlation, and it points to a structural reason, not leakage.
- **|num(x)| orders the miss better than den(x).** Spearman −0.96 vs −0.67 (order37). Slope $\hat{p} \approx 1.4$ is post hoc and was rightly not used.
- **The L1–L7 collision is still in hashed files.** order27 PREREG, BATCH8 PREREG and MF corrigenda still use L-labels. These are disclosed and allowed under CORE_FREEZE, and they need no action beyond the corrigenda.
- **“35 vs 36” usage is mostly consistent.** order36 states the distinction explicitly. The top-level and order20 wording “leaves the 35-component law vector span … vs 36” mixes the two terms, though it is not wrong.

Missed variables

variable	where written	never used as
Im $\hat{\kappa}$ (sign and size)	order20 (−4.8e−5i), order35 (−0.0229i), order36 (−0.0043i), order37 (−0.00034i), order31 (Im up to −0.16)	predictor or criterion; only $\lvert\hat{\kappa}-1\rvert$ is used
Q(edge) of the non-e1 edge	order34 “e2 … Q = 3.5”; order31 e1 Q = 1	regressor for r(T_B2) across cones
|num(x)| = den(−1/x), N′ = 2|num|	order37 diagnostics; L_B (order20)	preregistered predictor (order38 candidate)
max|coords| of cone	order31 Spearman +0.578	follow-up control
⟨γ,h⟩ character sectors (+,−), (−,+), (−,−) (ranks 24/18/27)	order27	any projection of S, X or the corner pieces
Π± sectors	order29 (Π+LN, Π−LN)	applied to T_e1 / T_BB / C2b
vtx share of each object	order30 lesson; order31 non-vacuity span	separate predictor
k = 0 share of S_in (0.325)	order35	control; order21 gives $c_0 = B2(\alpha 2)/N^2$
C2a, C11 for (0,1)-edge cones	order34 (identities for K only)	(0,1)-edge decomposition; order35 marks it out of

variable	where written	never used as
		scope
hit rate 1/9 (exact)	MARKOV_FIBRE_FREQ exact.py	weight for any Resonant comparison
u mod 3 independent of u mod 8/64 (MF3)	L3_INDEPENDENCE	joint mod-24 fibre variable
per-point leftover (median 0.46)	order36	only Spearman 0.18 is reported; not split by |num|

Patterns (not preregistered)

pattern	evidence
Edge ranking e1 ≺ e2 ≺ interior	order31, 32, 34: 0.003 / 0.022 / 0.04–0.17
Linear scale law T_e1(n) = ((2n+7)/4)C2b, i.e. B = 2n+7	order33/34; exact by definition. The e2 analogue was never written
r falls as grid fills (small-den points worst: 1, 2, ½, 3)	order35–37
Ordering by |num(x)|, slope ≈ 1.4	order37 (post hoc)
Fibre parity: C35 → odd fibres; (0,0) → {0,4}; MF2 odd mass ½	PST/PSM tables; MF2
Empty pairs differ by 4: {3,7} Residual, {2,6} T-phase	MF2 vs PSM1/PST1/PBR2
105-way splits: character ranks 36+24+18+27; C105 class; 105 companions	order27; PATTERN_ST_MARKOV
Defect directions scatter (mean |cos| 0.064 / 0.022)	order32 RG28; order33 RG30
Every predicted “cancellation” by subdivision diverges or stays orthogonal	order28–30

Correlations

pair	value	source
r vs den(x)	Spearman −0.67	order37
r vs |num(x)|	−0.96	order37
r vs arc length ℓ	Pearson −0.883 (mostly an e1 confound)	order31
S out-of-span vs X out-of-span	0.3678 vs 0.3677	order36
r vs |Im κ̃| across grids	co-monotone (3 grids)	order35–37
T_e1 vs C2b	exactly (7/4)·C2b, r equal to 2.5e−17	order34
r(C2b, P7) vs r(CO, full)	0.05853 vs 0.0585	order35 / order29

pair	value	source
MF2 empties vs Resonant (BATCH8) support	Resonant {0,2} ⊂ open; T-phase empties {2,6}	BATCH8, PST1
Edge type vs which piece carries the law	B2 part carries it on e1 only; T_BB fails on all 37	order33/34

Contradictions / tensions

#	claim A	claim B	severity
1	HUNT_ALPHA1: no written mod-8 T-phase indexing; maps paused	PBR2/PSM1/PST1 use T-phase mod 8, run afterwards, with no authorization line	**High**
2	order34: 0.3% miss “is the L_B leakage”	order35/36 FAIL both leakage measures	**High** (needs a forward note)
3	order30: near-law “looks generic”	order31: e1-only	Medium (superseded but not annotated)
4	BATCH8: written Resonant support {0,2}	T-phase/Markov pushes give fibre 2 mass 0	Medium (two “Resonant” fibre notions disagree)
5	order31: “four” e1 cones, criterion over 30	table has 5; NOTES “5/5 vs 0/26”	Low (count only)
6	PSM1, PBR2, PST1 reported as three FAILs	identical vector	Low/Medium (overcounting)
7	order35 r_meas(P7) = 0.0585 vs “full-grid 0.00305”	r(CO) = 0.0585 as well	Low (verify)
8	L1–L7 reserved	L-labels in order27 PREREG, BATCH8 PREREG	Low (corrigenda exist)
9	“35-component law vector span … vs 36” (order20/top-level)	order36: ambient rank 36 = dim V is a separate notion	Low (wording)
10	order33: Cor meets the criterion on 8/9 “K excluded as span-vacuous”	order31 counts K as satisfying	Low

Recommended next predefined tests (ranked by readiness)

1. **order38 RG35: Bernoulli-column reference span.** This is the written next step in order36/37 NOTES. Measure the leftover of S outside span{C2b_l(x), C2b_l(−1/x)} with no fit, against r. Uses stored arrays only.
2. **order38 secondary: |num(x)| monotone ordering.** Preregister a monotone decrease of the per-|num| median r with a fixed threshold and no fitted p (order37 NOTES). Add Im κ̂ per point as a reported co-variable.
3. **(0,1)-edge decomposition, RG31-style.** Use the order20 objects only: write the e2 analogue of T_e1 = (7/4)C2b and test r(T_e2) ≈ 0.022 with Q(e2) = 7/2 as the declared predictor. This applies the order34 identities to the second edge. Order35 marked it out of scope, so it needs a fresh PREREG.
4. (Bookkeeping, no test) Forward notes for #2 and #3. Merge PSM1, PBR2 and PST1 into one result. Confirm authorization for the four unlabelled PATTERN packages.
5. (Maps) **Stay paused for new fibres.** The parity/mod-4 pattern above may only become a test if the geometric side or Arvin writes the fibre definition, per HUNT_ALPHA1 (a)–(c).